

Cost Optimization design principles

Architecture design is always driven by business goals and must **factor in return on investment (ROI) and financial constraints**. Typical questions to consider include:

- Do the allocated budgets enable you to meet your goals?
- What's the spending pattern for the application and its operations? What are priority areas?
- How will you maximize the investment in resources, by better utilization or by reduction?

A cost-optimized workload isn't necessarily a low-cost workload. There are significant tradeoffs. Tactical approaches are reactive and can reduce costs only in the short term. To achieve long-term financial responsibility, you need to **create a strategy with prioritization, continuous monitoring, and repeatable processes** that focuses on optimization.

The design principles are intended to provide optimization strategies that you need to consider when you design and implement your workload architecture. Start with the recommended approaches and **justify the benefits for a set of business requirements**. After you set your strategy, drive actions by using the [Cost Optimization checklist](#) as your next step.

As you prioritize business requirements to align with technology needs, expect the initial cost allocation to change. However, you should expect a series of tradeoffs in areas in which you want to optimize cost, such as security, scalability, resilience, and operability. If the cost of addressing the challenges in those areas is high and these principles aren't applied properly, you might make risky choices in favor of a cheaper solution, ultimately affecting your organization's business goals and reputation.

Develop cost-management discipline

[Expand table](#)



Build a team culture that has awareness of budget, expenses, reporting, and cost tracking.

Cost optimization is conducted at various levels of the organization. It's important to understand how your workload cost is aligned with organizational FinOps practices. A view into the business units, resource organization, and centralized audit policies allows you to adopt a standardized financial system.

[Expand table](#)

Approach	Benefit
Develop a cost model. This fundamental exercise is a prerequisite to setting up a financial tracking system.	A cost model helps segment expenses and estimate and forecast the total cost of ownership, including infrastructure, support, and implementation. It enables you to identify cost drivers early and predict how any change, growth, or shrinkage will affect overall spending in your projected business model.
Have an effective but flexible accountability model that's governed and implemented with properly assigned roles and responsibilities.	Clear accountability helps enforce the functional expectations of each role (given a scope), drive clarity, and generate reports with transparency at desired levels. Proactive governance can help you avoid actions that might lead to unnecessary expenditure that's beyond the budget.
Estimate realistic budgets that cover all non-negotiable functional and nonfunctional requirements, personnel, and processes that provide for anticipated growth.	You'll be able to set financial boundaries and establish ways to check your spending against the allocated budget. You'll also get notifications when certain thresholds are exceeded, which prevents overspending at the tenant scope, resource scope, and other scopes that are applied to the budget.
For workloads governed by service-level agreements (SLAs), evaluate whether to allocate budget toward potential penalties or toward implementation efforts.	A well-implemented solution can help you avoid penalties altogether, making proactive investment. Setting aside a customer compensation budget is a pragmatic approach to reduce the risk of future liability. Collaborate with your product owner to negotiate realistic cost compensation budget.
Plan on training costs, hiring expenses, and the cost of infrastructure needed to augment skills as the workload matures.	Investing in staffing complements existing skills through full-time or vendor support.
Communicate cost implications of design changes that are driven by insights gained from production.	The organization is able to make practical budget adjustment based on production feedback, which should be considered as meaningful as numeric data.

Design with a cost-efficiency mindset

 Expand table



Spend only on what you need to achieve the highest return on your investments.

Every architectural decision has direct and indirect financial implications. Understand the costs associated with build versus buy options, technology choices, the billing model and licensing,

training, operations, and so on.

Given a set of requirements, optimize, and make tradeoff decisions, in relation to costs, that still effectively address the cross-cutting concerns of the workload.

 Expand table

Approach	Benefit
Establish a cost baseline, including the projected growth. Ensure design choices work within the allocated budget to meet the functional and nonfunctional requirements.	Cost estimates enable you to forecast expenses against the budget and pinpoint key cost drivers. They also help reveal hidden costs that might otherwise go unnoticed, supporting a balanced approach that avoids overengineering.
Factor in expenses related to technology choices, automation, acquisition, training, and change management as part of that total cost.	This process will also generate decision trees for technology options based on cost considerations. By eliminating high-cost alternatives that lack strong business justification, you can free up budget capacity to invest in higher-value opportunities. We don't recommend that you design beyond planned growth as it may result in a lower ROI.
Design and enforce cost guardrails in your architecture that keep the resources within the upper and lower limits.	Enforcement can prevent incidental or unapproved charges and ensure that only budgeted quantity of resources are provisioned.
Treat different SDLC environments differently, and deploy the right number of environments.	You can save money by understanding that not all environments need to simulate production. Nonproduction environments can have different features, SKUs, instance counts, and even logging. You also can save costs by creating preproduction environments on-demand and removing them when you no longer need them.

Design for usage optimization

 Expand table



Maximize the use of resources and operations. Apply them to the negotiated functional and nonfunctional requirements of the solution.

Services and offerings provide various capabilities and pricing tiers. After you purchase a set of features, avoid underutilizing them. Find ways to maximize your investment in the tier.

Likewise, continuously evaluate billing models to find those that better align to your usage, based on current production workloads.

 Expand table

Approach	Benefit
Take advantage of the full capabilities of your selected resource SKUs to meet performance, security, reliability, and operational goals.	You can maximize the use of what you paid for. Avoid selecting SKUs with features you don't need, as they can lead to unnecessary costs without added benefit.
Evaluate opportunities to dynamically adjust capacity, scaling up when demand increases and scaling down when it's no longer needed.	Without this approach, you may need to pre-provision more capacity than necessary. By contrast, dynamic scaling enables you to maintain a minimum baseline and expand only when required, aligning resource consumption with actual usage patterns.
Prioritize deployment of active-active models over active-passive models, as part of your recovery plan, if you already paid for the resources.	If your design defaults to using active-passive models, you might have idle resources that could otherwise be used. Converting to active-active might enable you to meet your load leveling and scale bursting requirements without overspending.
Prioritize the use of commitment-based discounted resources when developing new features, setting up additional environments, or optimizing for nonfunctional requirements.	Finding opportunities to use committed plans can significantly reduce the cost of implementing new functionality.
Make the most of your investment in a support plan. Keep allowance for training to ensure the team are using relevant tools and technologies.	Using your support plan to handle production problems or for proactive reviews will help you get your money's worth. Fully engage with your Microsoft support model.

Design for rate optimization

 Expand table



Increase efficiency without redesigning, renegotiating, or sacrificing functional or nonfunctional requirements.

Take advantage of opportunities to optimize the utility and costs of your existing resources and operations. If you don't, you unnecessarily spend money without any added ROI.

 Expand table

Approach	Benefit
Identify resources that have stable or predictable usage patterns over time. Optimize costs by prepurchasing these resources to take advantage of available discounts. Collaborate with your licensing team to influence future purchasing agreements and renewal strategies.	Microsoft offers discounted rates for predictable, long-term commitments to specific resources or resource categories. These resources incur lower costs during the usage period and can be amortized over time. By keeping your licensing team informed about current and projected resource investments, you can help them right-size commitments during agreement negotiations. In some cases, these forecasts can influence your organization's price sheet, benefiting not only your workload's cost efficiency but also other teams using the same technologies.
Explore alternatives that don't require additional licensing. Consider options like hybrid use and preproduction subscription pricing.	You'll be able to reduce licensing costs by taking advantage of options that give you usage rights to the same or comparable technologies at a lower cost.
Use consumption-based pricing when it's more cost effective.	You'll pay for what you use. This option might be more expensive than a fully utilized prepaid option. However, if you don't expect to fully utilize pre-purchased compute, pay-as-you-go might be a better choice.
Use fixed-price billing instead of consumption-based billing for a resource when its utilization is high and predictable and a comparable SKU or billing option is available.	When utilization is high and predictable, the fixed-price model usually costs less and often supports more features.
Where possible, co-locate usage with other workloads, resources, and teams to reduce financial and operational costs.	Shared resources are managed centrally and provisioned with higher capacity to support multiple workloads, allowing costs to be distributed across teams.
Deploy to lower-cost regions, provided there are no compromises to functional or nonfunctional requirements. Evaluate regional options for each environment individually. While production may require specific regions, consider leveraging more cost-effective regions for preproduction environments where feasible.	Use premium regions only where necessary can lead to significant savings. Also, savings from non-production environments can be reallocated to other priorities.
Prefer services that make it easier to achieve higher density.	As density increases, the amount of resources that you need to run a workload decreases. This decrease reduces cost per unit and the cost of management.

Approach	Benefit
Consider the potential tradeoffs, especially on security boundaries.	

Monitor and optimize over time

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Continuously right-size investment as your workload evolves with the ecosystem.

What was important yesterday might not be important today. As you learn through evaluation of production workloads, expect changes in architecture, business requirements, processes, and even team structure. Your software development lifecycle (SDLC) practices might need to evolve. External factors might also change, like the cloud platform, its resources, and your agreements.

You should carefully assess the impact of all changes on cost. Monitor changes and the ROI trend on a regular cadence, and evaluate whether you need to adjust functional and nonfunctional requirements.

[Expand table](#)

Approach	Benefit
Build capabilities in the system that capture and classify expense.	You'll be able to calculate the costs that reveal technical and business perspectives at different billing boundaries. You'll also be able to conduct regular reviews and drive showback and chargeback processes.
Implement cost alerts when spending approaches predefined budget thresholds. Regularly review and adjust these alerts to ensure they remain aligned with evolving usage patterns.	Proactive notifications help prevent budget overruns and support timely decision-making.
Continuously evaluate and adjust architecture design decisions around cost of resources, operations, and paid support.	Regular reviews of metrics, performance data, billing reports, and feature usage might lead to fine-tuning that can reduce costs. You might also be able to save some costs by evaluating the use of your support contract and right-sizing it.

Approach	Benefit
Decommission resources that are underutilized, unused, obsolete, or can be replaced with more efficient alternatives. Regularly delete unnecessary data.	By resizing or removing underutilized resources, or even changing SKUs, you can reduce costs. Shutting down unused resources and deleting data when you no longer need it reduces waste and frees up funds so you can invest them elsewhere.

Next steps

[Cost Optimization checklist](#)

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